

CSC 361

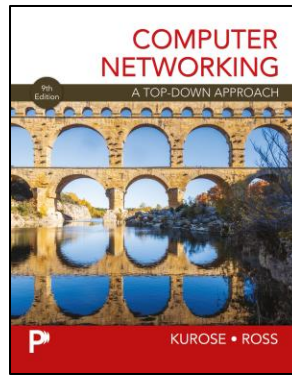
Computer Networks

Transport Layer

(Chapter 3 – TCP)

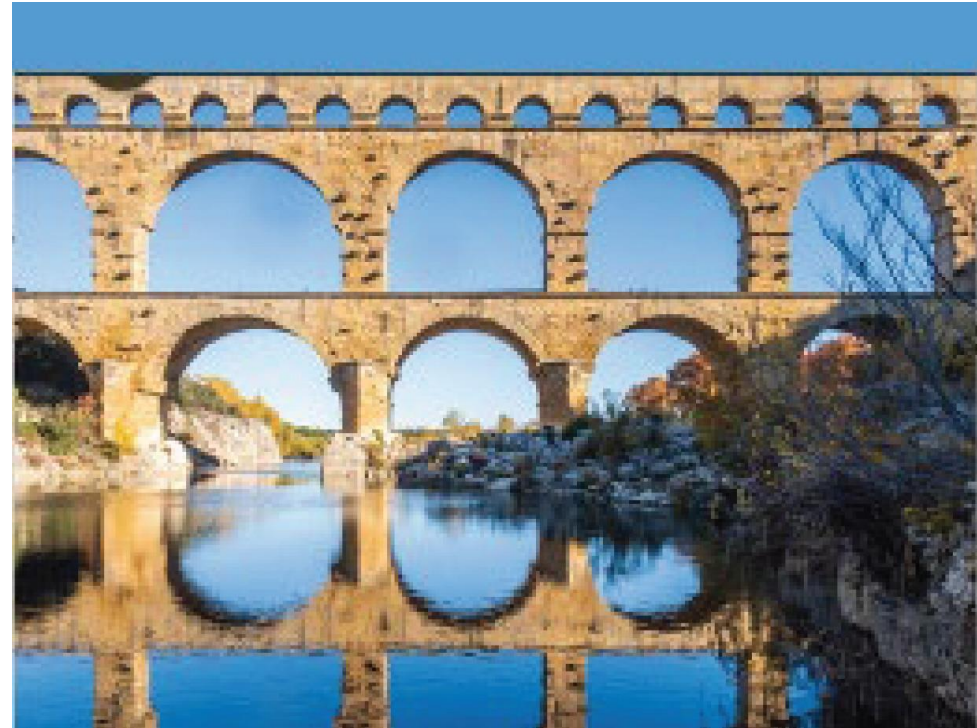
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Chapter 3: Roadmap (4 of 7)

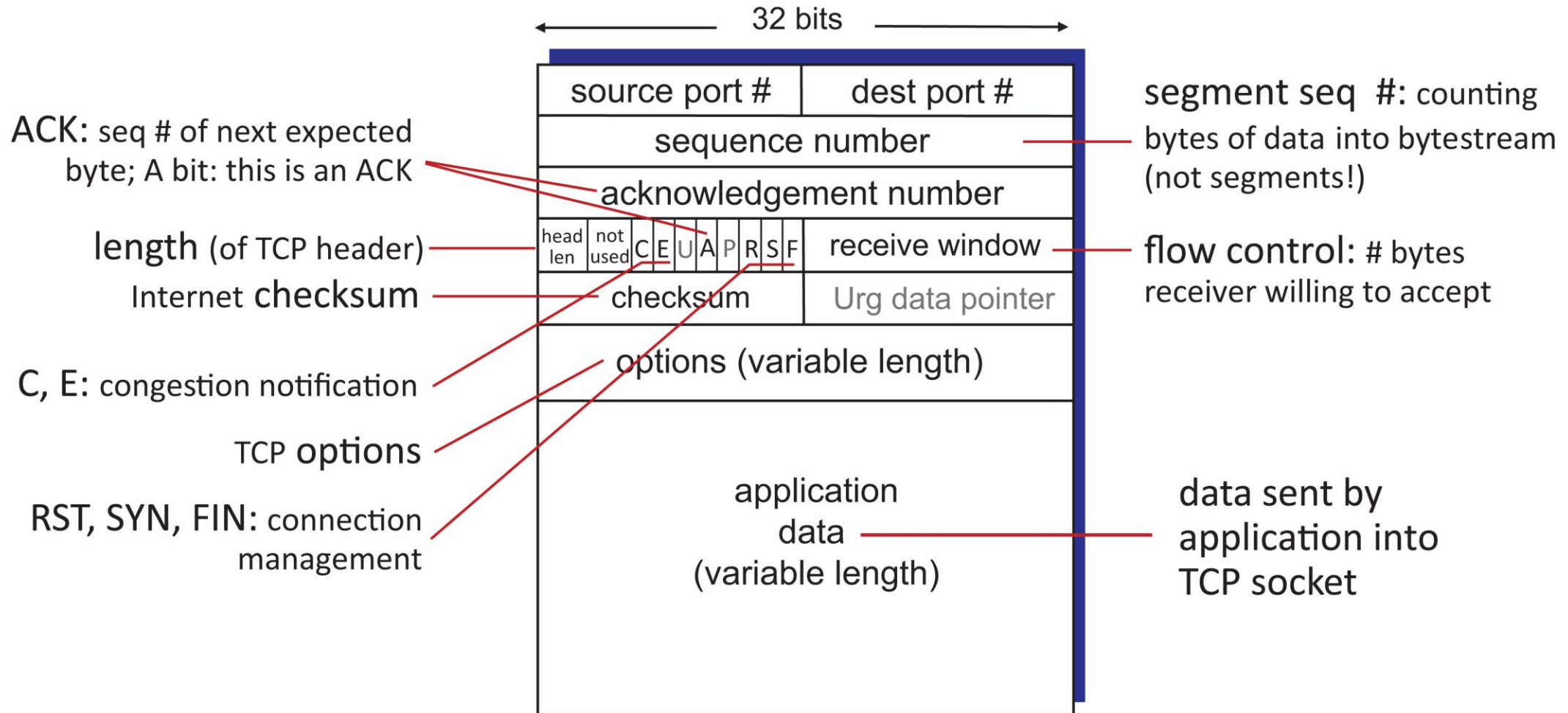
- Transport-layer services
- Multiplexing and demultiplexing
- Connectionless transport: UDP
- Principles of reliable data transfer
- **Connection-oriented transport: TCP**
 - segment structure
 - reliable data transfer
 - flow control
 - connection management
- Principles of congestion control
- TCP congestion control



TCP: Overview RFCs: 793, 1122, 2018, 5681, 7323

- **point-to-point:**
 - one sender, one receiver
- **reliable, in-order byte stream:**
 - no “message boundaries”
- **full duplex data:**
 - bi-directional data flow in same connection
 - MSS: maximum segment size
- **cumulative ACKs**
- **pipelining:**
 - TCP congestion and flow control set window size
- **connection-oriented:**
 - handshaking (exchange of control messages) initializes sender, receiver state before data exchange
- **flow controlled:**
 - sender will not overwhelm receiver

TCP Segment Structure



TCP Sequence Numbers, ACKs (1 of 2)

Sequence numbers:

- byte stream “number” of first byte in segment’s data

Acknowledgements:

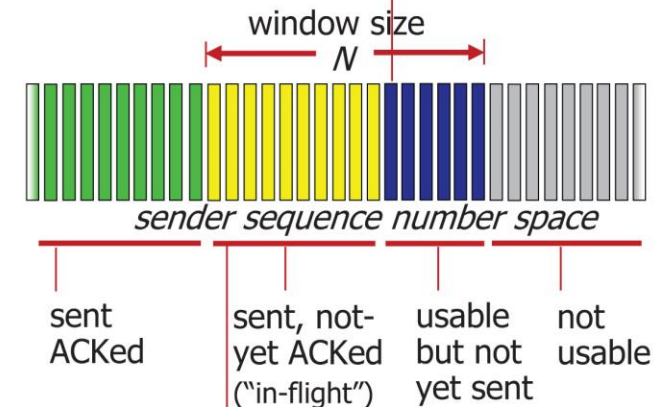
- seq # of next byte expected from other side
- cumulative ACK

Q: how receiver handles out-of-order segments

- **A:** TCP spec doesn’t say, – up to implementor

outgoing segment from sender

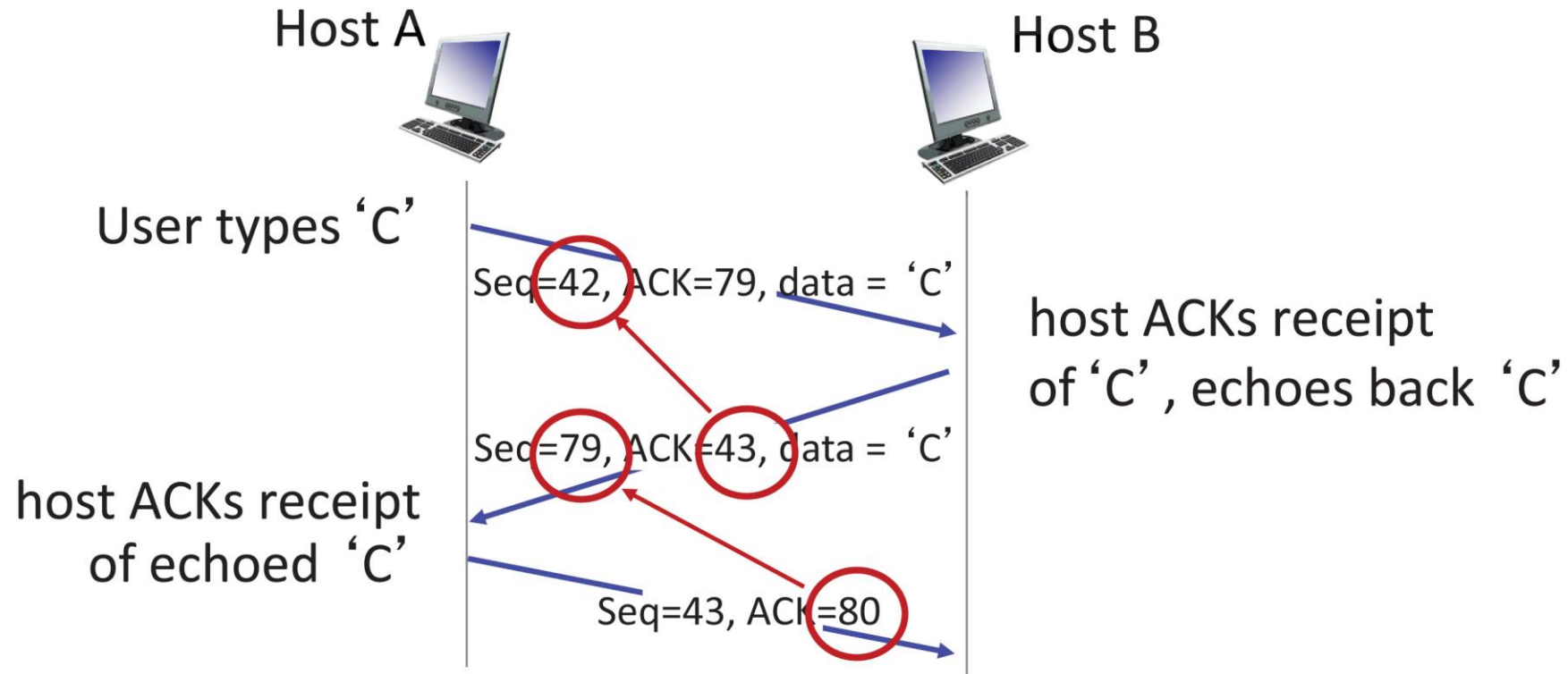
source port #	dest port #
sequence number	
acknowledgement number	
	rwnd
checksum	urg pointer



outgoing segment from receiver

source port #	dest port #
sequence number	
acknowledgement number	
	A
checksum	urg pointer

TCP Sequence Numbers, ACKs (2 of 2)



simple telnet scenario

TCP Round Trip Time, Timeout (1 of 3)

Q: how to set TCP timeout value?

- longer than RTT, but RTT varies!
- **too short:** premature timeout, unnecessary retransmissions
- **too long:** slow reaction to segment loss

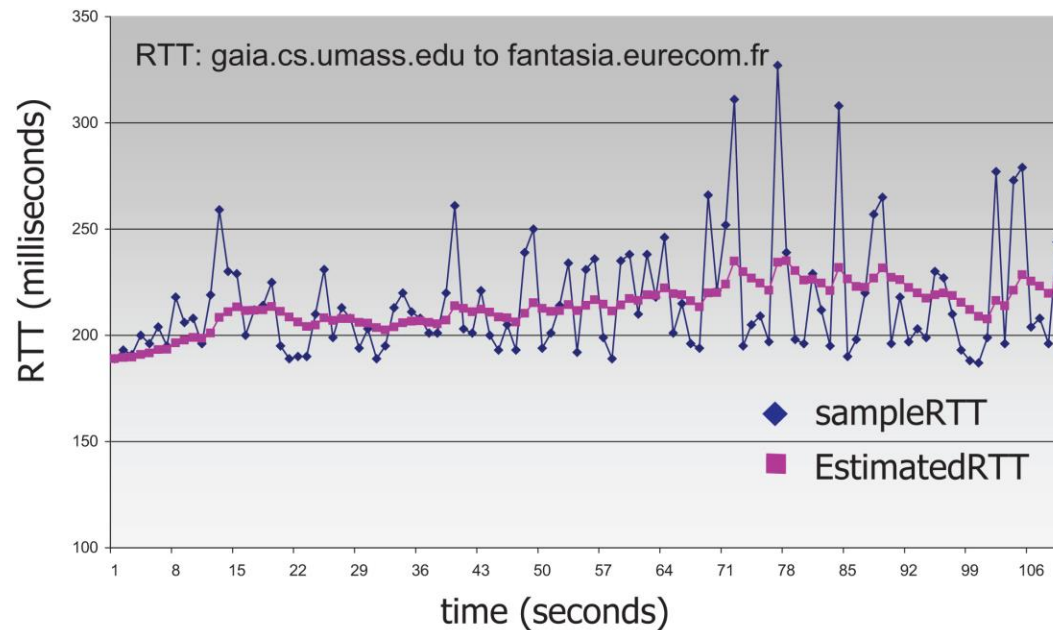
Q: how to estimate RTT?

- **SampleRTT:** measured time from segment transmission until ACK receipt
 - ignore retransmissions
- **SampleRTT** will vary, want estimated RTT “smoother”
 - average several **recent** measurements, not just current **SampleRTT**

TCP Round Trip Time, Timeout (2 of 3)

$$\text{EstimatedRTT} = (1 - \alpha) * \text{EstimatedRTT} + \alpha * \text{SampleRTT}$$

- exponential weighted moving average (EWMA)
- influence of past sample decreases exponentially fast
- typical value: $\alpha = 0.125$



TCP Round Trip Time, Timeout (3 of 3)

- timeout interval: **EstimatedRTT** plus “safety margin”
 - large variation in **EstimatedRTT**: want a larger safety margin

$$\text{TimeoutInterval} = \text{EstimatedRTT} + 4 * \text{DevRTT}$$



↑
estimated RTT

↑
“safety margin”

- **DevRTT**: EWMA of **SampleRTT** deviation from **EstimatedRTT**:

$$\text{DevRTT} = (1 - \beta) * \text{DevRTT} + \beta * |\text{SampleRTT} - \text{EstimatedRTT}|$$

(typically, $\beta = 0.25$)

* Check out the online interactive exercises for more examples:

http://gaia.cs.umass.edu/kurose_ross/interactive/

TCP Sender (Simplified) (1 of 2)

event: data received from application

- create segment with seq #
- seq # is byte-stream number of first data byte in segment
- start timer if not already running
 - think of timer as for oldest unACKed segment
 - expiration interval: **TimeoutInterval**

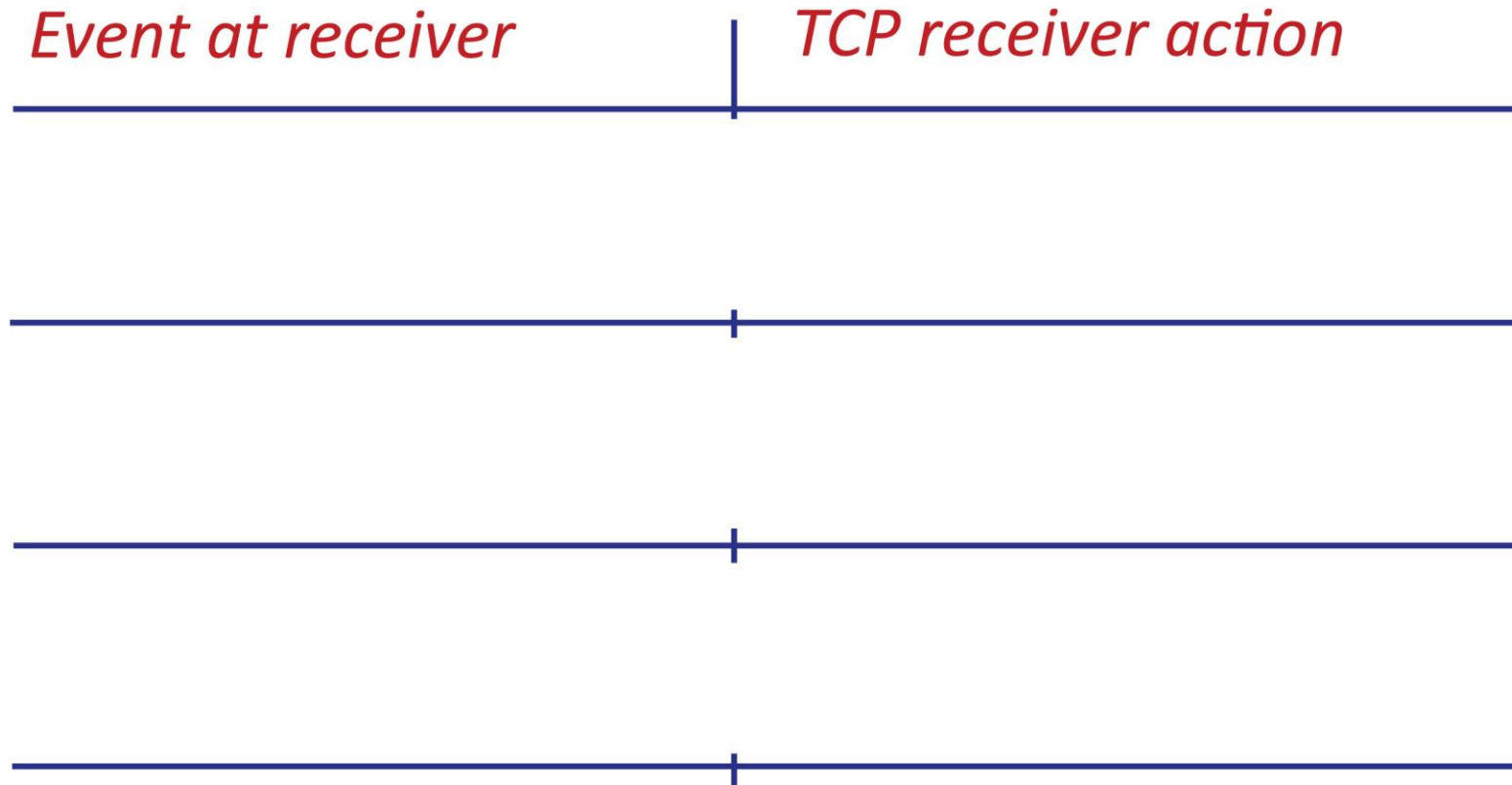
event: timeout

- retransmit segment that caused timeout
- restart timer

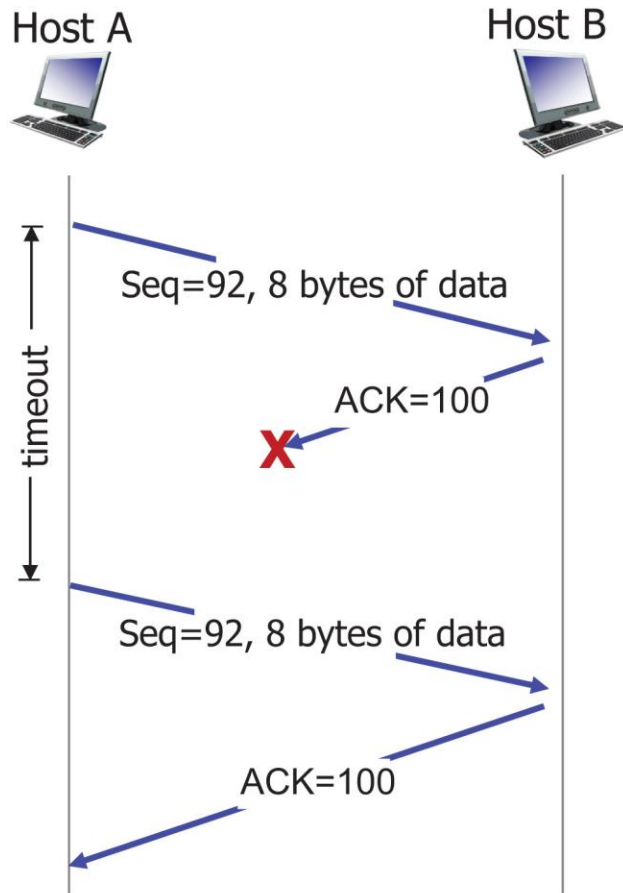
event: ACK received

- if ACK acknowledges previously unACKed segments
 - update what is known to be ACKed
 - start timer if there are still unACKed segments

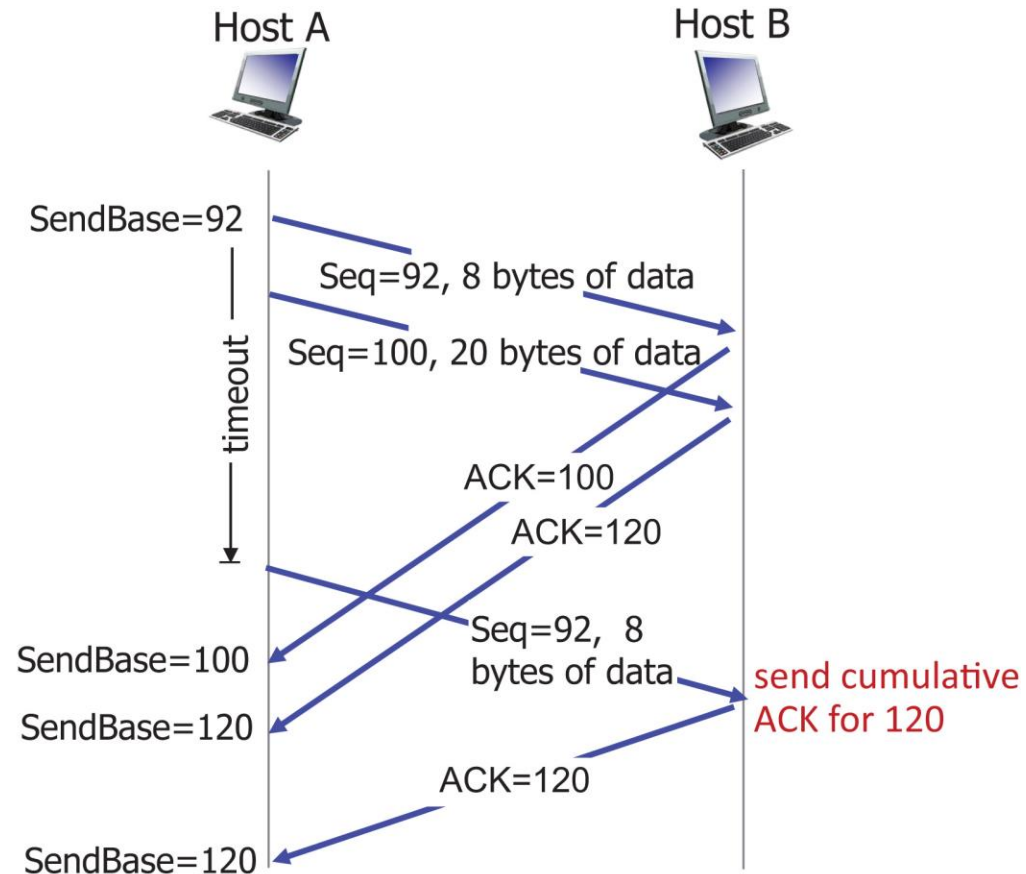
TCP Receiver: ACK Generation [RFC 5681]



TCP: Retransmission Scenarios (1 of 2)

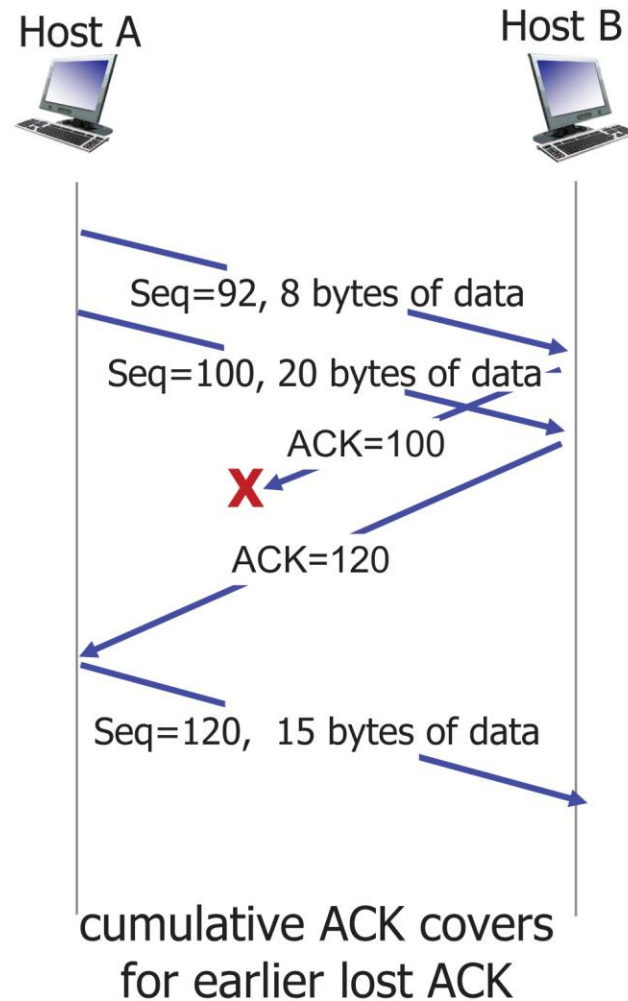


lost ACK scenario



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premature timeout
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TCP: Retransmission Scenarios (2 of 2)



TCP Fast Retransmit

TCP fast retransmit

if sender receives 3 additional ACKs for same data (“triple duplicate ACKs”), resend unACKed segment with smallest seq #

- likely that unACKed segment lost, so don't wait for timeout

💡 Receipt of three duplicate ACKs indicates 3 segments received after a missing segment – lost segment is likely. So retransmit!

